

TECHNICAL WHITE PAPER · v1.0

GravelKing Pro

A browser-native music-production platform: architecture, audio methodology, and honest limitations.

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Abstract

GravelKing Pro is a web-based music-production platform that unifies songwriting, audio separation, recording, mixing, mastering, and format conversion in a single application. This paper describes the system architecture, the Morris Law Kernel (MLK v3.5) audio engine, the methodology behind each tool, and — importantly — the honest limitations of the digital-signal-processing techniques used. Our design philosophy favors transparent, explainable processing over marketing claims the software cannot keep.

1. Introduction

Independent creators typically assemble a fragmented toolchain: a desktop DAW, a cloud stem-splitter, a mastering service, a converter, and a separate release platform. Each adds cost, friction, and data-handling risk. GravelKing Pro consolidates these into one browser application backed by a shared audio engine, with client-side processing where possible and first-party server processing otherwise.

2. Design principles

- Browser-first: the interactive DAW runs entirely client-side using the Web Audio API, avoiding upload latency and keeping project audio on the user's machine.
- First-party processing: server-side tools run on GravelKing infrastructure; user audio is never handed to third-party services.
- Explainable DSP: every tool states what it does and where it is approximate. We do not claim source separation or transcription we do not perform.
- Progressive entitlement: free tiers deliver real value; paid tiers unlock heavier compute and IP features.

3. System architecture

The platform is a pnpm monorepo. A Vite + React single-page application delivers the UI; an Express + TypeScript API handles processing, subscriptions, and the label/library. PostgreSQL (via Drizzle ORM) stores application state, and Stripe Checkout manages subscriptions. Audio work is split between the browser (Web Audio for the live DAW) and the server (ffmpeg-backed MLK v3 routines for splitting, mastering, and conversion).

Access to premium features is enforced server-side. Anonymous usage is tracked with a session cookie, and authenticated accounts use OpenID Connect; both are mapped to entitlements on the backend so client state can never unlock paid compute on its own.

4. The Morris Law Kernel (MLK v3.5)

MLK v3.5 is GravelKing's in-house DSP engine. It provides the carving and filtering primitives shared across tools — band-splitting, center-channel processing, EQ, dynamics, and loudness shaping. The kernel is CPU-oriented and tuned for deterministic, repeatable output rather than GPU acceleration. On

the server it is invoked through ffmpeg-native routines for throughput; in the browser the same DSP concepts are realized through Web Audio nodes.

A benchmark harness exercises the kernel with a 4096×4096 double-precision matrix multiply (DGEMM) over eight iterations to characterize throughput in GFLOPS. When the server kernel runner is unavailable, the benchmark interface presents clearly-labeled demonstration figures rather than failing silently.

I 5. Tool suite

5.1 Mixing Studio (live DAW)

A multi-track DAW running on the Web Audio API. Tracks can be recorded or imported; each supports volume, pan, and a chain of plugins (EQ, dynamics, time-based effects). Mixdown renders a stereo file in-browser. Projects can be serialized and restored.

5.2 Vocal Booth

Combines a teleprompter, a backing-track player, an optional guide-vocal stem, and a recorder. Users may split one of their own songs into instrumental + guide vocal for practice. The final mixdown deliberately contains only the instrumental and the user's recorded take — never the guide vocal.

5.3 Voice Splitter

Separates a track into instrumental and vocal by cancelling the center-panned content common to most stereo mixes.

5.4 Mastering

Applies a mastering chain — EQ, compression, loudness normalization, and optional denoise — via presets matched to genre.

5.5 Songwriting Studio

Generates complete songs from a prompt using Google's Gemini model. Paid users gain a line-by-line editor and an authorship-certification step.

5.6 File Converter

Transcodes audio and video into MP3, WAV, FLAC, M4A, or OGG using ffmpeg, extracting audio from video containers automatically.

5.7 Hardware Optimizer

Detects CPU topology and benchmarks audio-processing throughput, presented alongside the MLK optimization sequence.

6. Audio methodology and honest limitations

We consider transparency about limitations a core feature, not a disclaimer.

- Vocal separation is approximate. The Voice Splitter and Vocal Booth split rely on center-channel cancellation and band-splitting, not trained source separation. Expect midrange bleed; results are best on true stereo mixes and degrade on mono or heavily-bussed material.
- Lyric timing is energy-based. Teleprompter highlighting is derived from signal energy, not speech transcription or forced alignment, so line timing is approximate.
- The Gate plugin behaves as a downward compressor rather than a sample-accurate noise gate; the other EQ and dynamics plugins follow standard RBJ biquad formulations.
- Benchmark figures may be demonstrative. When the server kernel runner is unavailable, the optimizer shows labeled demo numbers rather than fabricated live measurements.

7. AI songwriting and authorship

The Songwriting Studio uses generative AI to draft lyrics and structure. We are explicit that free AI drafts carry no intellectual-property rights to the user. Paid users perform meaningful line-by-line editing and then certify their human–AI authorship, producing a record of human contribution; finished works can be exported to Suno for audio generation. This positions AI as a co-writer with a clear, documented human-authorship trail.

8. Privacy and data handling

The live DAW processes audio entirely in the browser, so project audio need not leave the user's device. Server-side tools process on GravelKing's own infrastructure and do not forward user audio to third-party processors. Subscription and billing data are handled through Stripe; webhooks are signature-verified against the raw request body before any state change.

9. Performance and hardware optimization

The platform targets commodity CPUs. The optimization sequence illustrates the kernel's tuning concepts — core affinity and memory locking — while the DGEMM harness provides a throughput reference. We intentionally present optimization steps as an illustrative walkthrough rather than guaranteed changes to a user's local machine, since server-side processing characteristics differ from a client device.

10. Subscriptions and access model

GravelKing Pro offers a free Starter tier, a weekly plan, a monthly Pro Plus plan, and an enterprise Node Auditor plan, each with server-enforced entitlements. Paid plans include free-trial periods and a 30-day money-back guarantee. Entitlement checks live on the API so that no client-side flag can unlock paid compute.

11. Roadmap

- Trained neural source separation for higher-fidelity stems, offered as a premium cloud-GPU path.
- A sample-accurate true noise gate in the DAW plugin set.
- Expanded authorship-certification exports and label-distribution tooling.

12. Conclusion

GravelKing Pro demonstrates that a serious, end-to-end music-production workflow can live in the browser without sacrificing honesty about its methods. By unifying creation, processing, and release behind a shared DSP engine — and by stating plainly where techniques are approximate — the platform earns trust while lowering the cost and friction of making music.